# Experiment 16:- Write a Program to Implement K-fold Validation Technique of Machine Learning on any Dataset

In [1]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.svm  import SVC
from  sklearn.ensemble import RandomForestClassifier
import numpy as np

from sklearn.datasets import  load_digits
digits = load_digits()
```

In [2]:
```
from sklearn.model_selection  import train_test_split
x_train,x_test,y_train,y_test = train_test_split(digits.data,digits.target,test_s
```

In [3]:
```
Ir = LogisticRegression()
Ir.fit(x_train,y_train)
Ir.score(x_test,y_test)
```

```
C:\dev\code\python ml\python_experiments_ipynb\.venv\Lib\site-packages\sklearn\lin
ear_model\_logistic.py:599: ConvergenceWarning: lbfgs failed to converge after 100
iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=100).
You might also want to scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

Out[3]:
```
0.912962962962963
```

In [4]:
```
svm = SVC()
svm.fit(x_train,y_train)
svm.score(x_test,y_test)
```

Out[4]:
```
0.9537037037037037
```

In [5]:
```
from sklearn.model_selection import  KFold
kf = KFold(n_splits=3)
kf
```

Out[5]:
```
KFold(n_splits=3, random_state=None, shuffle=False)
```

In [6]:
```
for train_index ,test_index in kf.split([1,2,3,4,5,6,7,8,9]):
    print(train_index,test_index)
```

```
[3 4 5 6 7 8] [0 1 2]
[0 1 2 6 7 8] [3 4 5]
[0 1 2 3 4 5] [6 7 8]
```

In [7]:
```
def get_score(model,x_train,x_test,y_train,y_test):
    model.fit(x_train,y_train)
    return model.score(x_test,y_test)
```

In [8]:
```
get_score(LogisticRegression(),x_train,x_test,y_train,y_test)
#get_score(SVC(),x_train,x_test,y_train,y_test)|
#get_score(rf,x_train,x_test,y_train,y_test)
```

```
C:\dev\code\python ml\python_experiments_ipynb\.venv\Lib\site-packages\sklearn\lin
ear_model\_logistic.py:599: ConvergenceWarning: lbfgs failed to converge after 100
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    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

Out[8]: 0.912962962962963